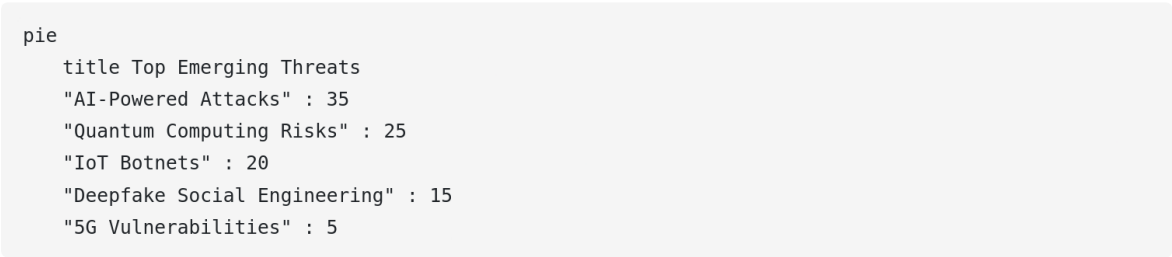


Cybersecurity Training Diary

Day 21: Emerging Threats & Future Trends

1. Topics Covered

✓ 2024 Threat Landscape



✓ Cutting-Edge Defenses

Technology	Application	Example Tools
AI Security	Anomaly detection	Darktrace, IBM Watson for Cyber
Post-Quantum Crypto	Quantum-resistant algorithms	CRYSTALS-Kyber (NIST Standard)
Deception Tech	Fake assets to trap attackers	IllusionBLACK, TrapX

✓ Career Specializations

- Threat Intelligence Analyst
- Cloud Security Architect
- Red Team Operator
- ICS/OT Security Engineer

2. Key Takeaways

- ✓ AI is a double-edged sword: Both attackers and defenders are adopting it
- ✓ Quantum computers will break RSA/ECC: Start planning crypto agility now
- ✓ Skills of the future: Cloud security, AI governance, automation

3. Practical Exercises

♦ Exercise 1: Simulate AI-Powered Phishing

```
# GPT-3 phishing email generator (hypothetical)
import openai
response = openai.ChatCompletion.create(
    model="gpt-4",
    messages=[{"role": "user", "content": "Write a CEO fraud email requesting urgent wire transfer"}]
)
print(response.choices[0].message.content)
```

♦ Exercise 2: Analyze Quantum Risks

```
# Check OpenSSL for post-quantum algorithms
openssl list -signature-algorithms | grep -i "dilithium"
```

4. Questions for Reflection

- ? How should organizations prepare for quantum threats?
- ? What ethical concerns arise with AI in cybersecurity?